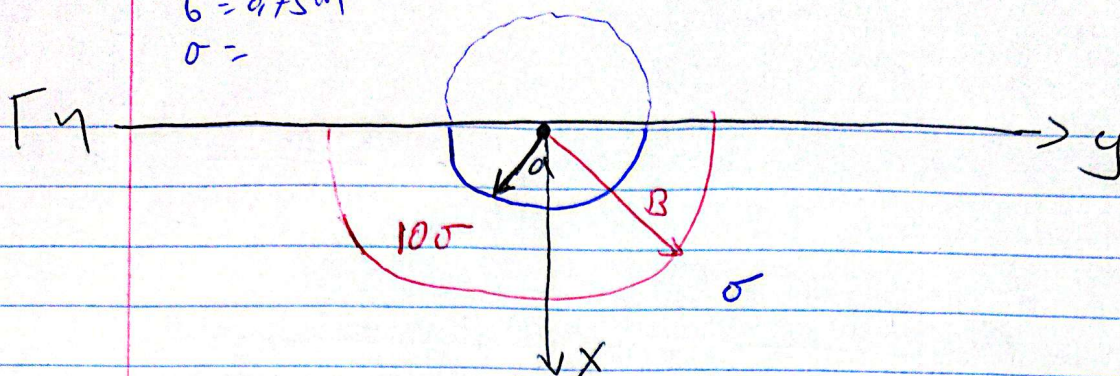


$$a = 0.5\text{m}$$

$$b = 0.75\text{m}$$

$$\sigma =$$



Λν I το πεδίο, έστω 2I ολόκληρης σφαίρας, τότε

$$2I = \int_0^{2\pi} \int_0^{\pi} |\vec{J}| \cdot r_0^2 \cdot \sin(\theta) d\theta d\phi$$

$$2I = |\vec{J}| \cdot r^2 \cdot 4\pi = 2I \Rightarrow |\vec{J}| = \frac{I}{2\pi r^2}$$

$$\vec{J} = \frac{I}{2\pi r^2} \hat{r}_0$$

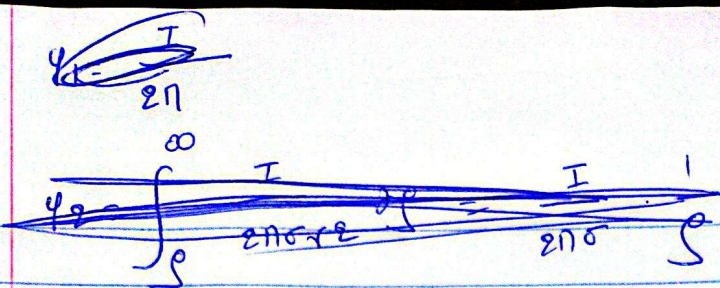
$$\vec{E}(\rho) = \begin{cases} \frac{I}{2\pi 100 r^2} & , a \leq r \leq b \\ \frac{I}{2\pi \sigma r^2} & , B < r < \infty \end{cases}$$

$$\varphi_L = \int_{-\infty}^a 0 d\rho + \int_a^B \frac{I}{2\pi 100 r^2} d\rho + \int_B^{\infty} \frac{I}{2\pi \sigma r^2} d\rho$$

$$\frac{I}{2\pi 100} \left[\frac{1}{a} - \frac{1}{B} \right] + \frac{I}{2\pi \sigma} \left[\frac{1}{B} \right]$$

$$\text{Για } r \leq a \Rightarrow \frac{I}{2\pi 100} \left[\frac{1}{a} - \frac{1}{B} \right] + \frac{I}{2\pi \sigma} \cdot \frac{1}{B}$$

$$\varphi_L = \frac{I}{2\pi \sigma} \left[\frac{1}{10a} - \frac{1}{10B} + \frac{1}{B} \right] \Rightarrow \varphi_L = \frac{I}{2\pi \sigma} \left[\frac{1}{10a} + \frac{9}{10B} \right]$$



$$V_2 = \int_a^B \frac{I}{2\pi\epsilon_0 r^2} dr + \int_B^\infty \frac{I}{2\pi\epsilon_0 r^2} dr$$

$$V_2 = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{r} - \frac{1}{B} \right] + \frac{I}{2\pi\epsilon_0 B} \quad a \leq r \leq B$$

$$V_3 = \int_B^\infty \frac{I}{2\pi\epsilon_0 r^2} dr \Rightarrow V_3 = \frac{I}{2\pi\epsilon_0 B} \quad r \geq B$$

$$V_{AB} = V(a) - V(B) = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{9}{10B} \right] - \frac{I}{2\pi\epsilon_0} \left[\frac{1}{B} - \frac{1}{B} \right] + \frac{I}{2\pi\epsilon_0 B}$$

$$V_{AB} = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{9}{10B} + \frac{10}{10B} \right] \Rightarrow V_{AB} = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{19}{10B} \right]$$

$$V_{B\infty} = V(B) - V(\infty)$$

$$V_{B\infty} = \frac{I}{2\pi\epsilon_0 B} - \frac{I}{2\pi\epsilon_0 \infty} \Rightarrow V_{B\infty} = \frac{I}{2\pi\epsilon_0 B}$$

$$V_0 = V_{AB} + V_{B\infty} = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{19}{10B} \right] + \frac{I}{2\pi\epsilon_0 B}$$

$$V_0 = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{29}{10B} \right]$$

$$\text{As per } V_0 = \frac{I}{2\pi\epsilon_0} \left[\frac{1}{10a} + \frac{9}{10B} \right] \Rightarrow I = \frac{2\pi\epsilon_0 \cdot V_0}{\frac{1}{10a} + \frac{9}{10B}}$$

$$R_{0L} = V_{AB} \cdot I, \quad R_{0R} = V_{B\infty} \cdot I$$